

Machine learning for the prediction of first-year mortality in incident hemodialysis patients

--Manuscript Draft--

Manuscript Number:	EBIOM-D-20-01353
Article Type:	Article (Original Research)
Keywords:	Machine learning; hemodialysis; XGBoost; prediction model
Corresponding Author:	Jianghua Chen the First Affiliated Hospital, College of Medicine, Zhejiang University Hangzhou, CHINA
First Author:	Kaixiang Sheng
Order of Authors:	Kaixiang Sheng Xi Yao Jiawei Li Yongchun He Ping Zhang Jianghua Chen
Manuscript Region of Origin:	CHINA
Abstract:	Rationale The current mortality risk scores for hemodialysis (HD) patients employ regression techniques and have limited applicability and robustness. Objective We aimed to develop a machine learning model utilizing clinical factors to predict first-year mortality in HD patients that could assist physicians in classifying high-risk patients. Methods Training and testing cohorts consisted of 5351 incident HD patients from a single center and 5828 incident HD patients from 97 renal centers, respectively. The outcome was all-cause mortality during the first year of dialysis. eXtreme Gradient Boosting (XGBoost) was used for algorithm training and validation. Two models were established based on the data obtained at dialysis initiation (model 1) and data from 0-3 months after dialysis initiation (model 2). Ten-fold cross-validation was applied to each model. The C statistic (area under the receiver operating characteristic curve, AUC) was used to assess the predictive ability of the models. Results In the training and testing cohorts, 585 (10.93%) and 764 (13.11%) patients, respectively, died during the first-year follow-up. Fifteen most important features were selected from 42 candidate features. The performance of model 1 (C statistic of 0.8318, 95% CI 0.7771 - 0.8374) was similar to that of model 2 (C statistic of 0.8530, 95% CI 0.8057 - 0.8593). Conclusions We developed and validated two machine learning models to predict first-year mortality in HD patients. These models could be used to stratify high-risk patients in the early stages of dialysis.

Machine learning for the prediction of first-year mortality in incident hemodialysis patients

Kaixiang Sheng¹ • Xi Yao¹ • Jiawei Li¹ • Yongchun He¹ • Ping Zhang¹ • Jianghua Chen¹

1. Kidney Disease Center, the First Affiliated Hospital, College of Medicine, Zhejiang University; Key Laboratory of Kidney Disease Prevention and Control Technology, Zhejiang Province; The Third Grade Laboratory under the National State, Administration of Traditional Chinese Medicine, Hangzhou China 310003.

Correspondence to Jianghua Chen (zjukidney@zju.edu.cn) and Ping Zhang (1194076@zju.edu.cn):

Address: Kidney Disease Center, the First Affiliated Hospital College of Medicine Zhejiang University, Qingchun Road 79, Hangzhou, 310003, China.

Phone: +8657187236992

Fax number: +8657187236189

Abstract

Rationale: The current mortality risk scores for hemodialysis (HD) patients employ regression techniques and have limited applicability and robustness.

Objective: We aimed to develop a machine learning model utilizing clinical factors to predict first-year mortality in HD patients that could assist physicians in classifying high-risk patients.

Methods: Training and testing cohorts consisted of 5351 incident HD patients from a single center and 5828 incident HD patients from 97 renal centers, respectively. The

outcome was all-cause mortality during the first year of dialysis. eXtreme Gradient Boosting (XGBoost) was used for algorithm training and validation. Two models were established based on the data obtained at dialysis initiation (model 1) and data from 0-3 months after dialysis initiation (model 2). Ten-fold cross-validation was applied to each model. The C statistic (area under the receiver operating characteristic curve, AUC) was used to assess the predictive ability of the models.

Results: In the training and testing cohorts, 585 (10.93%) and 764 (13.11%) patients, respectively, died during the first-year follow-up. Fifteen most important features were selected from 42 candidate features. The performance of model 1 (C statistic of 0.8318, 95% CI 0.7771 - 0.8374) was similar to that of model 2 (C statistic of 0.8530, 95% CI 0.8057 - 0.8593).

Conclusions: We developed and validated two machine learning models to predict first-year mortality in HD patients. These models could be used to stratify high-risk patients in the early stages of dialysis.

Keywords: machine learning; hemodialysis; XGBoost; prediction model

Introduction

The overall prevalence of chronic kidney disease (CKD) is 10.8% in China and 15% in the United States, which has brought a significant economic, social, and medical burden on patients and society.^[1-3] According to the United States Renal Data System, there are approximately 0.12 million patients with end-stage renal disease (ESRD) starting chronic renal replacement therapy every year.^[4] However, survival among incident hemodialysis (HD) patients remains poor, especially in the first year

of the initiation of dialysis.^[5, 6]

It is essential for us to stratify the risk for mortality according to the clinical and laboratory findings of HD patients. The identification of HD patients who are at high risk of first-year mortality is of great clinical significance. It can inform patients of their survival prognosis in the early stages of dialysis and allow clinicians to make targeted intervention strategies to improve first-year outcomes.

Although some clinical risk prediction models have been developed for early mortality in the dialysis population, most of them are based on linear model (logistic or cox model).^[7-12] The performances of these models were not good enough in either the original population or the external validation, and none of these models' C-statistics is more than 0.8.

In recent years, machine learning has been proven to be a very powerful method by researchers in medical fields.^[13-16] Machine learning is useful in identifying the most important factors and developing predictive models with the best performance. A recent study reported a machine learning model to predict the first year survival of incident HD patients by using random forest. The model's C-statistics was 0.7504 (95% confidence intervals 0.7425–0.7583), which is superior to those of traditional risk prediction models.^[17] However, this is not accurate enough for an actual application.

In this study, we sought to build a prediction model based on machine learning techniques, utilizing readily available clinical factors to predict first-year mortality in incident dialysis patients.

Methods

Study Population

Adult ESRD patients (aged ≥ 18 years) with follow-up exceeding 12 months who started maintenance HD were included. Patients who died within 12 months of

follow-up were also included.

The exclusion criteria were as follows: patients with a history of previous renal replacement therapy; patients recovered kidney function within 3 months; patients who received renal transplantation or switched to peritoneal dialysis within 12 months after dialysis initiation. We also excluded patients with missing information on age at dialysis initiation or disease diagnoses.

Training data were retrieved consecutively from the First Affiliated Hospital College of Medicine Zhejiang University between January 2007 and April 2019 (Figure 1). Testing data were collected from 97 renal centers between January 2010 and August 2018 (Figure 1) for external validation. All follow-up data were updated to August 2017.

This study followed the tenets of the Declaration of Helsinki and was approved by the ethics committee of the First Affiliated Hospital of Zhejiang University (IIT20200088A), Hangzhou, China. Written informed consent was obtained for all study participants.

Outcome and predictors

The outcome of this study is all-cause mortality during the first year of dialysis. Outcome status and potential candidate variables for the prediction tool, including demographic information, disease diagnoses, comorbid information, and laboratory test results, were obtained from the Zhejiang Dialysis System.

Demographic information and type of vascular access were collected at the start of dialysis. Disease diagnoses, comorbid information, and laboratory test results were collected from 0-3 months after dialysis initiation. The most recent serum creatinine

measurements prior to the index date were used to estimate the glomerular filtration rate (GFR) using the CKD-EPI (Chronic Kidney Disease Epidemiology Collaboration) equation.^[18]

A total of 42 variables were included for candidate features, and only body mass index (BMI) and ferritin had missing rates less than 6% (Table 1). Before the baseline model was developed, missing data were imputed with the mean value for continuous variables and the most common value for categorical variables. By using one-hot encoding, all categorical features were transformed into numerical features. Box-Cox transformation was performed to normalize numerical features that were highly skewed.^[19]

Predictive Modeling

A gradient boosting machine learning algorithm (eXtreme Gradient Boosting, XGBoost) was employed to build a model to predict the correlation between parameters and the outcome. XGBoost is an integrated learning algorithm based on gradient boosted decision trees (GBDT).^[20] By using XGBoost, we estimated the feature importance scores of candidate features after going through the training process. The feature importance scores showed how valuable each feature was in the construction of the boosted decision trees within the model, which was calculated based on the Gini impurity index.^[21]

The XGBoost algorithm was employed for the following reasons: (1) it has high efficiency and accuracy, (2) it can prevent overfitting via regularization, (3) it provides feature importance, (4) and lastly, it allows the use of a wide variety of

computing environments.

Other popular machine learning algorithms, adaptive boosting (Adaboost), light gradient boosting machine (LightGBM), logistic regression (LR), random forest (RF), extra-trees (ET), gradient boosting (GB), multiple layers perception (MLP), k-nearest neighbor (KNN), and decision tree (DT) were also applied for comparison with XGBoost.

Two models were established based on the data obtained at dialysis initiation (model 1) and data from 0-3 months after dialysis initiation (model 2). Ten-fold cross-validation was used to avoid overfitting and to validate each model.^[22] Finally, the C statistic (area under the receiver operating characteristic curve, AUC) was used to assess the predictive ability of each model.

The Shapley Additive exPlanations (SHAP) was used to measure the marginal contribution of each feature to the models and explain the XGBoost results.^[23]

Results

Basic statistics

The demographic and clinical characteristics of the training and testing cohorts, which indicate that most characteristics were similarly distributed between the two cohorts, are shown in Table 1. The mean ages at dialysis initiation were 51.67 years in the training cohort and 62.53 years in the testing cohort. 61.58% of the patients in the training cohort and 60.47% in the testing cohort were men. All of the patients were Chinese. 585 (10.93%) deaths were reported in the training cohort and 764 (13.11%) deaths in the testing cohort.

Model Performance

Figure 2 shows the rank of features selected after training the XGBoost models. The same 15 most important features were chosen for the construction of model 1 and model 2.

Among the 11 algorithms applied, the XGBoost algorithm had the best performance in both models. The prediction performances of the two XGBoost models are shown in Table 2. The performances in model 1 (C statistic of 0.8318, 95% CI 0.7771 - 0.8374) and model 2 (C statistic of 0.8530, 95% CI 0.8057 - 0.8593) were similar (Figure 3).

The SHAP value results showed how features contributed to the prediction models (Supplementary Figure S1). It also explained how features computed the probability for a single participant. Figure 4 shows an example of a random participant using model 2. This participant had history of congestive heart failure, low hemoglobin and creatinine levels, high neutrophil and C-reactive protein levels, and old age at dialysis initiation, which contributed to a higher probability of mortality in the first year, although he had normal BMI and slightly high systolic blood pressure levels.

Discussion

The mortality of HD patients during the first year of dialysis initiation is high.^[5] Therefore, early and precise individualized risk estimates are required for clinical decision-making. However, it is hard to predict outcomes in HD patients because of the heterogeneity of primary disorders and broad comorbidities. Traditional strategies for building prediction models have contributed to quality improvement and decision support. Nevertheless, these models have some limitations that may lead to missing

important predictors and relationships.

In this study, by implementing advanced machine learning techniques, we developed and validated two clinical risk prediction models for first-year mortality in incident HD patients. Our prediction models, compared with previous models, were more accurate in stratifying the risk of first-year mortality for HD patients. Our prediction models have several unique and important characteristics. First, many clinical features have been reported for the prediction of first-year mortality in incident HD patients; some of these features are interact with each other. Traditional prediction models do not account for interactions between input features. By using XGBoost, our study selected the 15 most important features from 42 candidate features, and then combined them in a non-linear and highly interactive way. Second, missing data and data 'noise' are inevitable in clinical data collected from the real-world, which is a complex problem for traditional strategies. Machine learning techniques can deal with missing data and data 'noise' automatically to improve model performance. Third, relationships between data may change over time because of the improvements in treatment and changing populations. For example, the rates of diabetic nephropathy and cardiovascular disease have been increasing year after year. Traditional prediction models are always non-renewable. Machine learning allows for the continual update of the model to incorporate new data and capture changes in the relationships between features. We plan to display these prediction models on the website of the Zhejiang Dialysis Quality Control Center and as a mobile application for better application. Finally, compared with traditional predictive models, machine

learning models are more complex and hard to interpret; it is not easy to figure out how these models make decisions. Therefore, we used SHAP to increase the interpretability of the models in this study. What is more, SHAP results for a single patient can help physicians evaluate prognosis and make individualized treatment regimens.

Some previous studies used data obtained at dialysis initiation to predict the 3-6 months mortality of HD patients.^[11, 24] Floege et al., by using 90- to 180-day baseline and 0- to 90-day baseline data for the prediction of first-year mortality, revealed that models had similar performances.^[10] In our study, two models were established based on the data available at dialysis initiation and data from 0-3 months after dialysis initiation. The performance of model 1 (C statistic 0.8318) was similar to that of model 2 (C statistic 0.8530), suggesting that we can predict first-year mortality in HD patients at dialysis initiation. High-risk patients can be classified and treated with preventive therapies at the early stage of dialysis.

Despite the promising prospect of our study, it has some limitations. First, our training data is based on retrospective data generated from a single center. Therefore, a possible center effect cannot be excluded. Second, although no restriction was placed on ethnicity, all HD patients included were Chinese. The primary disease of ESRD and cardiovascular conditions of HD patients in China differ from those of HD patients in other regions.^[2, 25] Thus, the applicability of our models to other ethnic groups and regions needs to be confirmed. Third, we only assessed 1-year mortality, whereas long-term mortality is also important.^[26] Therefore, we plan to establish a

model to predict 2-year and 5-year mortality in our future study. Finally, therapeutic intervention data, such as the dialysis dose and frequency, were not used in this study. Because therapeutic interventions were not always fixed until 1-2 months after dialysis initiation, and therapeutic interventions in patients varied.

In conclusion, to accurately predict first-year mortality in incident HD patients, we developed two machine learning models based on the data available at dialysis initiation and data from 0-3 months after dialysis initiation. The algorithms were well-calibrated, and the overall diagnostic performances of the two models were similar.

Acknowledgements:

Support: This work is supported by National Key Research and Development Projects of China (2018YFC1314003). Study sponsors had no role in study design; collection, analysis, and interpretation of data; writing the report; and the decision to submit the report for publication.

Financial Disclosure: The authors declare that they have no other relevant financial interests.

Author contributions: Research idea and study design: KXS, PZ; data acquisition: KXS, XY; data analysis/interpretation: KXS, XY; statistical analysis: KXS, JWL; supervision or mentorship: JHC. Each author contributed important intellectual content during manuscript drafting or revision and accepts accountability for the overall work by ensuring that questions pertaining to the accuracy or integrity of any

portion of the work are appropriately investigated and resolved.

Disclosure Statement

There are no conflicts of interest to declare.

Reference:

1. Zhang L, Wang F, Wang L, *et al.* Prevalence of chronic kidney disease in China: a cross-sectional survey. *Lancet* 2012, **379**(9818): 815-822.
2. Saran R RB, Abbott KC, *et al.* US Renal Data System 2018 Annual Data Report: Epidemiology of Kidney Disease in the United States. *American journal of kidney diseases : the official journal of the National Kidney Foundation* 2019, **73**(3S1): A7-A8.
3. Collaboration GBDCKD. Global, regional, and national burden of chronic kidney disease, 1990-2017: a systematic analysis for the Global Burden of Disease Study 2017. *Lancet* 2020.
4. Saran R, Robinson B, Abbott KC, *et al.* US Renal Data System 2018 Annual Data Report: Epidemiology of Kidney Disease in the United States. *American journal of kidney diseases : the official journal of the National Kidney Foundation* 2019, **73**(3S1): A7-A8.
5. Robinson BM, Zhang J, Morgenstern H, *et al.* Worldwide, mortality risk is high soon after initiation of hemodialysis. *Kidney international* 2014, **85**(1): 158-165.

6. Soucie JM, McClellan WM. Early death in dialysis patients: risk factors and impact on incidence and mortality rates. *Journal of the American Society of Nephrology : JASN* 1996, **7**(10): 2169-2175.
7. Mauri JM, Cleries M, Vela E, *et al.* Design and validation of a model to predict early mortality in haemodialysis patients. *Nephrology, dialysis, transplantation : official publication of the European Dialysis and Transplant Association - European Renal Association* 2008, **23**(5): 1690-1696.
8. Chua HR, Lau T, Luo N, *et al.* Predicting first-year mortality in incident dialysis patients with end-stage renal disease - the UREA5 study. *Blood purification* 2014, **37**(2): 85-92.
9. Doi T, Yamamoto S, Morinaga T, *et al.* Risk Score to Predict 1-Year Mortality after Haemodialysis Initiation in Patients with Stage 5 Chronic Kidney Disease under Predialysis Nephrology Care. *PloS one* 2015, **10**(6): e0129180.
10. Floege J, Gillespie IA, Kronenberg F, *et al.* Development and validation of a predictive mortality risk score from a European hemodialysis cohort. *Kidney international* 2015, **87**(5): 996-1008.
11. Wick JP, Turin TC, Faris PD, *et al.* A Clinical Risk Prediction Tool for 6-Month Mortality After Dialysis Initiation Among Older Adults. *American journal of kidney diseases : the official journal of the National Kidney Foundation* 2017, **69**(5): 568-575.
12. Quinn RR, Laupacis A, Hux JE, *et al.* Predicting the risk of 1-year mortality in incident dialysis patients: accounting for case-mix severity in studies using

- administrative data. *Medical care* 2011, **49**(3): 257-266.
13. Attia ZI, Kapa S, Lopez-Jimenez F, *et al.* Screening for cardiac contractile dysfunction using an artificial intelligence-enabled electrocardiogram. *Nature medicine* 2019, **25**(1): 70-74.
 14. Chen T, Li X, Li Y, *et al.* Prediction and Risk Stratification of Kidney Outcomes in IgA Nephropathy. *American journal of kidney diseases : the official journal of the National Kidney Foundation* 2019.
 15. He J, Baxter SL, Xu J, *et al.* The practical implementation of artificial intelligence technologies in medicine. *Nature medicine* 2019, **25**(1): 30-36.
 16. Topol EJ. High-performance medicine: the convergence of human and artificial intelligence. *Nature medicine* 2019, **25**(1): 44-56.
 17. Akbilgic O, Obi Y, Potukuchi PK, *et al.* Machine Learning to Identify Dialysis Patients at High Death Risk. *Kidney international reports* 2019, **4**(9): 1219-1229.
 18. Levey AS, Stevens LA, Schmid CH, *et al.* A new equation to estimate glomerular filtration rate. *Annals of internal medicine* 2009, **150**(9): 604-612.
 19. Asar Ö, İlk O, Dag O. Estimating Box-Cox power transformation parameter via goodness-of-fit tests[J]. *Communications in Statistics-Simulation and Computation*, 2017, **46**(1): 91-105.
 20. Chen T GC. XGBoost: a scalable tree boosting system. *In Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* 2016: 785–794.

21. T. Hastie, R. Tibshirani, and J. Friedman, The Elements of Statistical Learning, 12th printing, Springer Series in Statistics. New York, NY, USA: Springer New York Inc., 2017.
22. Molinaro AM, Simon R, Pfeiffer RM. Prediction error estimation: a comparison of resampling methods. *Bioinformatics* 2005, **21**(15): 3301-3307.
23. Lundberg SM LS. A unified approach to interpreting model predictions. In: Advances in Neural Information Processing Systems; 2017.
24. Couchoud CG, Beuscart JB, Aldigier JC, *et al.* Development of a risk stratification algorithm to improve patient-centered care and decision making for incident elderly patients with end-stage renal disease. *Kidney international* 2015, **88**(5): 1178-1186.
25. Wang F, Yang C, Long J, *et al.* Executive summary for the 2015 Annual Data Report of the China Kidney Disease Network (CK-NET). *Kidney international* 2019, **95**(3): 501-505.
26. Arase H, Yamada S, Hiyamuta H, *et al.* Modified creatinine index and risk for long-term infection-related mortality in hemodialysis patients: ten-year outcomes of the Q-Cohort Study. *Scientific reports* 2020, **10**(1): 1241.

Titles and legends:

Figure 1: A workflow to develop the prediction models for first-year mortality in incident hemodialysis patients.

Figure 2: The feature importance ranking of XGBoost (with 42 features) based on

data at dialysis initiation (A) and data between 0-3 months after dialysis initiation (B).

CGN: chronic glomerulonephritis, DN: diabetic nephropathy, HTN: hypertensive nephropathy, LN: lupus nephritis, AAV: ANCA-associated vasculitis, GouN: gouty nephropathy, PKD: polycystic kidney disease, MM: multiple myeloma, AF: atrial fibrillation, CHF: congestive heart failure, COPD: chronic obstructive pulmonary disease, CVA: cerebrovascular disease, IHD: ischemic heart disease, BMI: body mass index, SBP: systolic blood pressure, DBP: diastolic pressure, Cr: creatine, BUN: urea nitrogen, UA: uric acid, Alb: albumin, eGFR: estimated glomerular filtration rate, P: phosphorus, Ca: calcium, K: potassium, Hb: hemoglobin, NEU: neutrophil, Plt: platelet, TG: triglycerides, TC: cholesterol, HDL: high-density lipoprotein, LDL: low-density lipoprotein, VLDL: very low-density lipoprotein, CRP: C-reactive protein, PTH: parathyroid hormone.

Figure 3: Receiver-operating characteristic curves of model 1 and model 2.

Figure 4: The SHAP value for a single data sample.

Table 1: Baseline characteristics of the training and testing cohorts

	Training cohort (at dialysis initiation, n=5351)	Testing cohort (at dialysis initiation, n=5828)	Training cohort (0-3 months, n=4425)	Testing cohort (0-3 months, n=3729)
Sex, male, %	3295 (61.58%)	3524 (60.47%)	2744 (62.01%)	2264 (60.71%)
Body mass index (kg/m ²)*	22.09±3.29	21.73±3.07	22.19±3.39	21.83±3.04
Age at dialysis initiation (years)	51.67±16.48	62.53±16.20	52.61±16.59	62.45±15.9
Systolic pressure (mmHg)	137.49±22.93	146.18±24.58	138.52±23.15	146.33±24.68
Diastolic pressure (mmHg)	77.76±12.26	78.95±15.52	80.45±12.15	79.02±15.45
Chronic kidney disease etiology, %				
Chronic glomerulonephritis	2823 (52.76%)	3015 (51.73%)	2445 (55.25%)	2064 (55.35%)
Diabetic nephropathy	1120 (20.93%)	1191 (20.44%)	895 (20.23%)	818 (21.94%)
Hypertensive nephropathy	262 (4.90%)	557 (9.56%)	218 (4.93%)	370 (9.92%)
Lupus nephritis	68 (1.27%)	50 (0.86%)	57 (1.29%)	29 (0.78%)
ANCA-associated vasculitis	57 (1.07%)	64 (1.10%)	53 (1.20%)	33 (0.88%)
Gouty nephropathy	32 (0.60%)	125 (2.14%)	26 (0.59%)	72 (1.93%)
Polycystic kidney disease	286 (5.34%)	214 (3.67%)	220 (4.97%)	150 (4.02%)
Other	703 (13.14%)	612 (11.07)	511 (11.54%)	204 (5%)
Comorbid conditions, %				
Cirrhosis	86 (1.61%)	90 (1.54%)	81 (1.83%)	60 (1.61%)
Multiple myeloma	46 (0.86%)	90 (1.54%)	46 (1.04%)	51 (1.37%)
Atrial fibrillation	108 (2.02%)	109 (1.87%)	85 (1.92%)	72 (1.93%)
Congestive heart failure	969 (18.11%)	999 (17.14%)	794 (17.94%)	605 (16.22%)
Ischemic heart disease	1476 (27.58%)	1578 (27.08%)	1206 (27.25%)	983 (26.36%)
Metastatic cancer	86 (1.61%)	91 (1.56%)	74 (1.67%)	38 (1.02%)
Lymphoma	7 (0.13%)	7 (0.12%)	6 (0.14%)	1 (0.03%)
Chronic obstructive pulmonary disease	241 (4.50%)	165 (2.83%)	169 (3.82%)	78 (2.09%)
Cerebrovascular disease	322 (6.02%)	411 (7.05%)	244 (5.51%)	271 (7.27%)
Laboratory data				
Leukocyte (10 ⁹ /L)	7.32±2.95	7.71±3.79	7.40±3.09	6.90±3.22
Neutrophil (10 ⁹ /L)	5.23±2.68	5.06±3.32	5.36±2.78	4.22±2.57
Hemoglobin	94.82±23.30	83.09±19.12	91.05±21.68	86.50±14.67
Platelet	193.28±93.47	182.47±83.70	190.84±88.13	184.36±71.39
Albumin	36.01±6.75	33.27±5.99	36.80±6.59	33.98±5.54
Phosphorus(mmol/L)	1.81±0.62	1.70±0.66	1.66±0.52	1.54±0.5
Calcium (mmol/L)	2.15±0.28	2.02±0.30	2.14±0.22	2.08±0.23

Potassium(mmol/L)	4.87±1.11	4.52±0.91	4.76±0.96	4.42±0.69
Parathyroid hormone (pg/ml)	334.71±292.07	246.95±193.61	315.98±291.84	241.26±206.48
Creatine (μmol/L)	807.11±352.04	718.84±336.47	755.28±315.95	661.5±268.48
Urea nitrogen (mmol/L)	22.65±12.07	23.61±11.77	19.87±8.72	20.01±8.13
Uric acid(μmol/L)	436.84±147.54	450.27±157.44	392.87±126.48	402.19±113.46
C-reactive protein	40.84±44.09	25.65±44.46	18.52±35.01	20.23±31.22
Cholesterol (mmol/L)	4.34±1.30	4.30±1.42	4.27±1.23	4.34±1.25
Triglycerides (mmol/L)	1.56±1.00	1.60±1.03	1.58±0.96	1.63±0.97
High-density lipoprotein (mmol/L)	1.14±0.42	1.11±0.43	1.12±0.39	1.15±0.38
Low-density lipoprotein (mmol/L)	2.36±1.10	2.37±1.02	2.31±1.04	2.35±0.92
Very low-density lipoprotein (mmol/L)	1.65±1.55	2.11±1.35	1.63±1.54	1.60±0.93
Ferritin (ng/mL)**	174.59±126.34	328.25±295.78	144.34±144.87	305.42±278.73
eGFR (mL/min/1.73m ²)	6.75±3.79	7.28±3.93	7.23±3.85	7.58±3.44
Vascular access at dialysis initiation, %				
Non-tunneled catheter	3295 (61.58%)	3388 (58.13%)	2495 (56.38%)	1893 (50.76%)
Tunneled catheter	1068 (19.96%)	1266 (21.72%)	1005 (22.71%)	938 (25.15%)
Fistula or graft	988 (18.46%)	1174 (20.14%)	925 (20.90%)	898 (24.08%)
Death at 1-year follow-up, %	585 (10.93%)	764 (13.11%)	437 (9.88%)	477 (12.79%)

*: The missing rates of body mass index in four cohorts were 5.04%,5.11%, 4.74% and 4.50%, respectively.

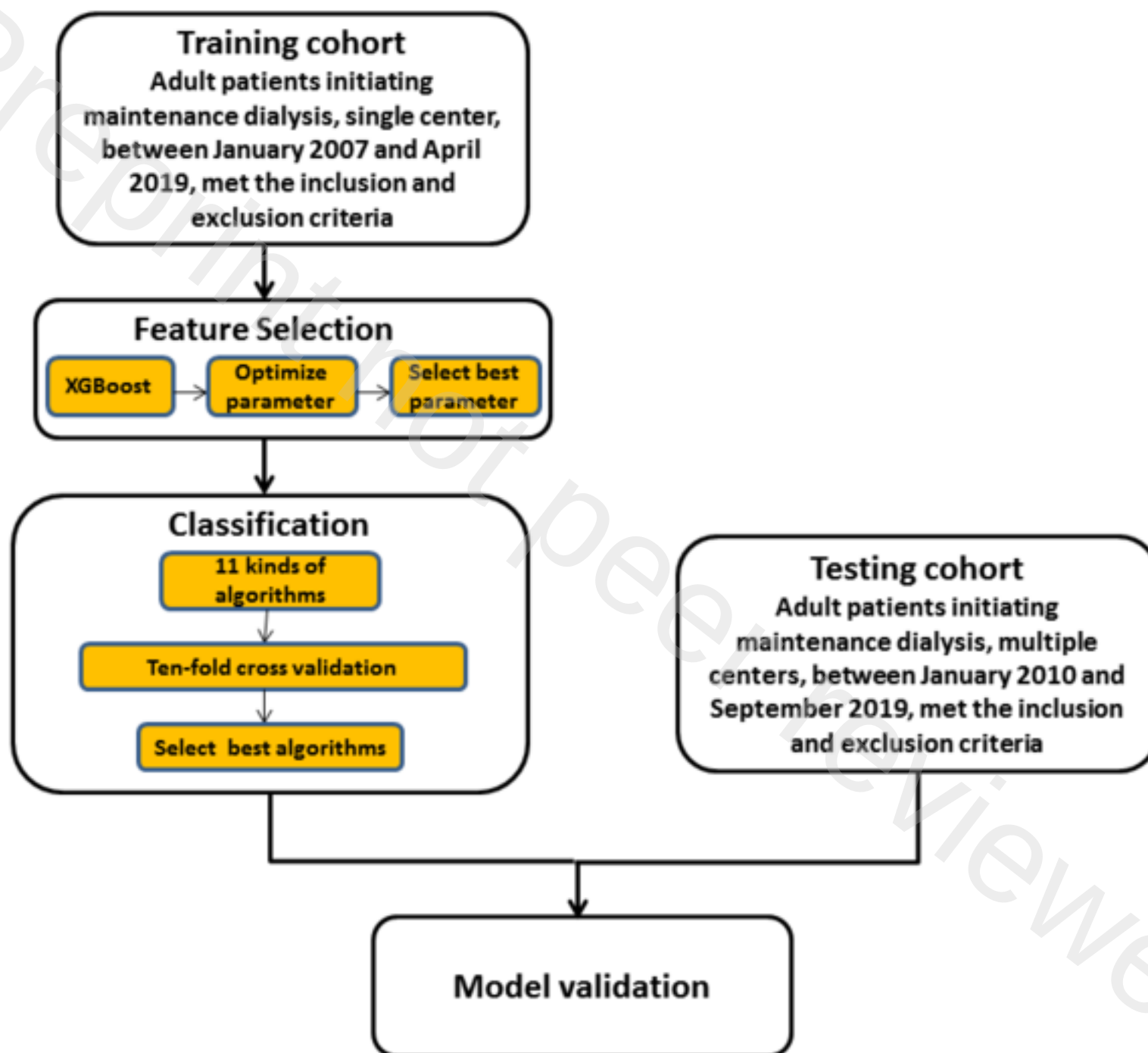
**:The missing rates of ferritin in four cohorts were 0.36%, 3.00%, 0.36% and 2.13%, respectively.

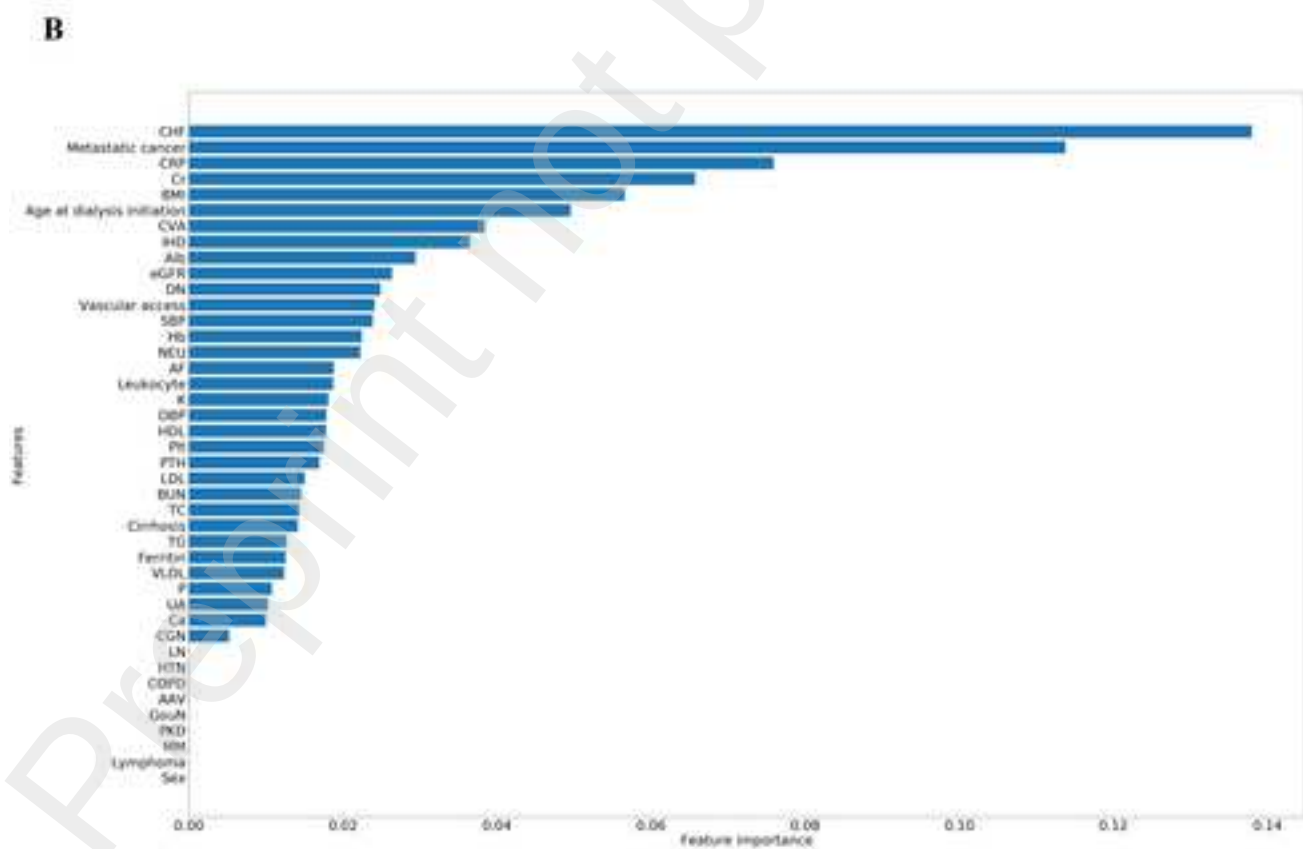
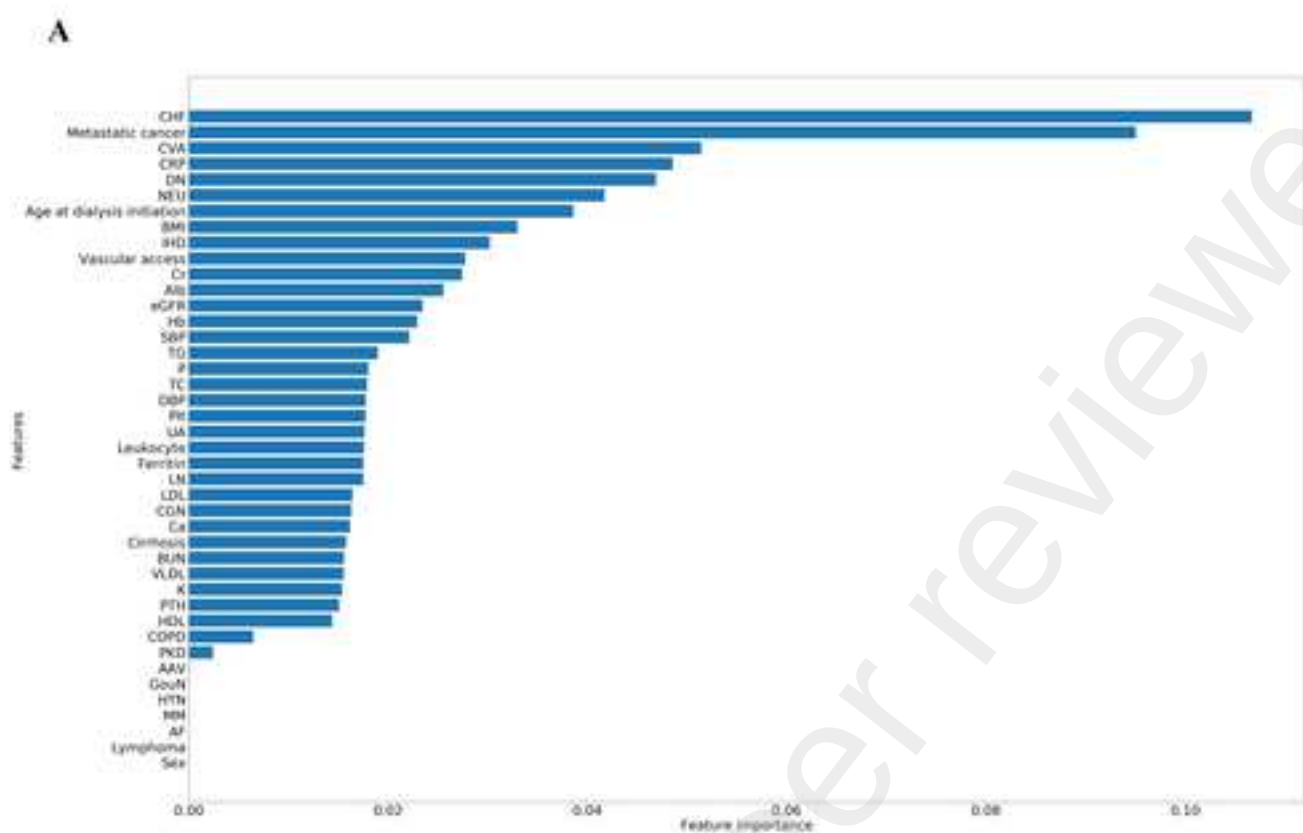
Abbreviation: eGFR = estimated glomerular filtration rate.

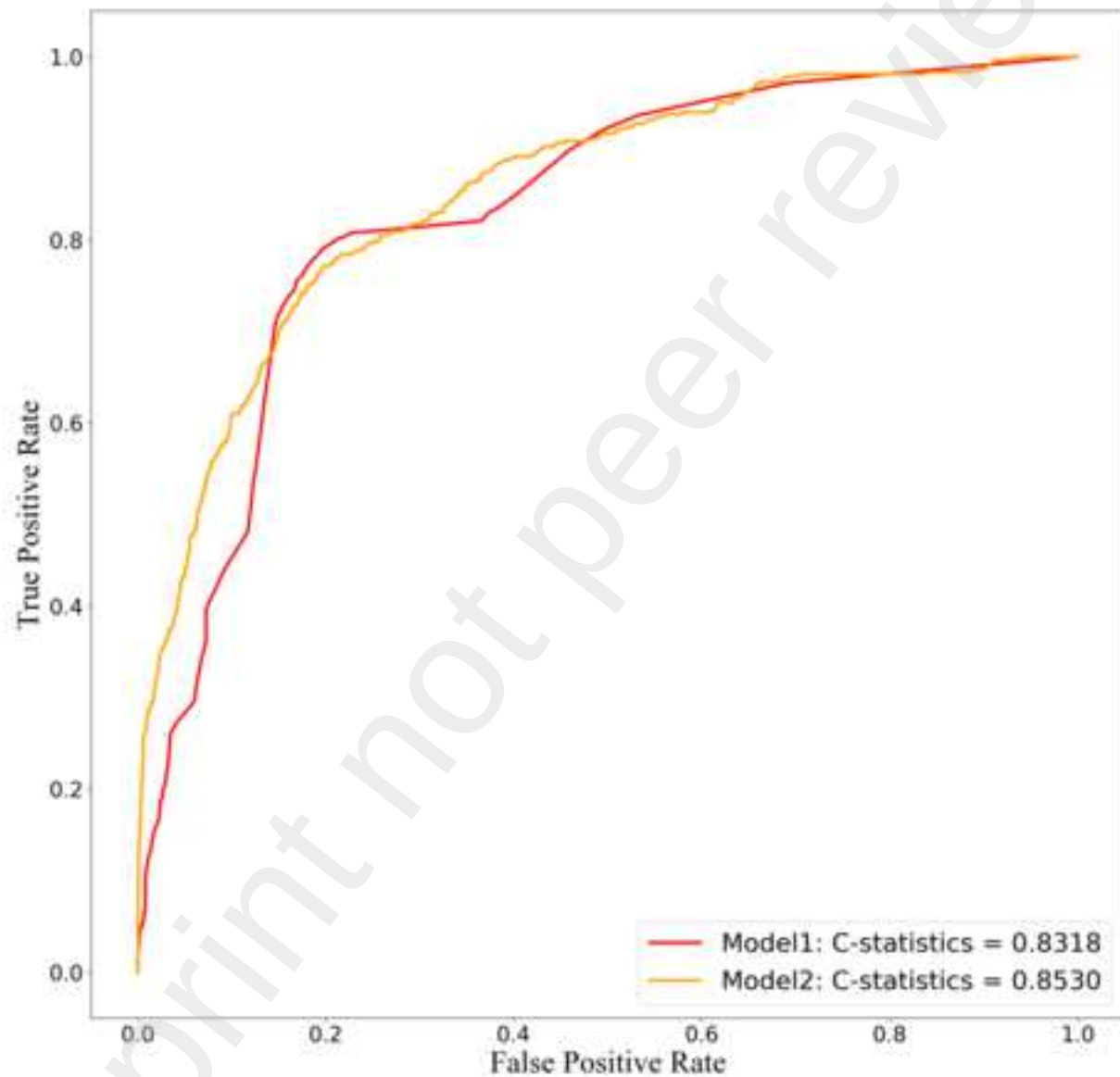
Table 2: C Statistics results of different models in the testing cohorts

	C Statistic	95% confidence interval	Accuracy
Model 1			
XGBoost	0.8318	0.7771 – 0.8374	85.50%
ET	0.8256	0.7709 – 0.8331	87.46%
LightGBM	0.8247	0.7745 – 0.8345	86.93%
GB	0.8177	0.7726 – 0.8326	80.22%
AdaBoost	0.8121	0.7686 – 0.8226	84.15%
RF	0.8087	0.7758 – 0.8337	87.35%
MLP	0.7981	0.6795 – 0.8355	47.19%
DT	0.7809	0.7648 – 0.7964	87.70%
KNN	0.7596	0.7577 – 0.8367	86.89%
LDA	0.7458	0.7437 – 0.8361	33.15%
LR	0.6795	0.6795 – 0.8355	14.91%
Model 2			
XGBoost	0.8530	0.8057 – 0.8593	87.24%
LightGBM	0.8471	0.8048 – 0.8548	87.88%
ET	0.8362	0.8013 – 0.8478	88.92%
GB	0.8336	0.8032 – 0.8485	87.42%
AdaBoost	0.8300	0.7996 – 0.8422	84.79%
KNN	0.8212	0.8056 – 0.8585	87.21%
RF	0.8199	0.8049 – 0.8538	77.69%
DT	0.8104	0.7966 – 0.8249	86.56%
LDA	0.7573	0.7557 – 0.8574	21.86%
LR	0.7288	0.7287 – 0.8567	19.87%
MLP	0.7147	0.7140 – 0.8561	13.41%

Abbreviation: XGBoost: eXtreme Gradient Boosting, Adaboost: adaptive boosting, LightGBM: light gradient boosting machine, LR: logistic regression, RF: random forest, ET: extra-trees, GB: gradient boosting, MLP: multiple layers perception, KNN:k-nearest neighbor, DT: decision tree.









Click here to access/download
e-component
Supplementary Material.docx

[Click here to access/download](#)

e-component

[Approve letter of clinical research ethics committee.pdf](#)

Click here to access/download
e-component
Informed consent form.docx